

Lab 02: Feature Engineering

✓ Example 01: Normalization and **Scaling**

```
from sklearn import datasets
import numpy as np
from sklearn import preprocessing

iris = datasets.load_iris()

X = iris.data[:, [2, 3]]
y = iris.target

std_scale = preprocessing.StandardScaler().fit(X)
X_std = std_scale.transform(X)

minmax_scale = preprocessing.MinMaxScaler().fit(X)
X_minmax = minmax_scale.transform(X)
```

✓ Example 02: Univariate **Analysis**

```
from sklearn import datasets
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

iris = datasets.load_iris()

# Let's convert to dataframe
iris = pd.DataFrame(
    data=np.c_[iris['data'], iris['target']],
    columns=iris['feature_names'] + ['species']
)

# replace the values with class labels
iris.species = np.where(
    iris.species == 0.0,
    'setosa',
    np.where(iris.species == 1.0, 'versicolor', 'virginica')
)

# let's remove spaces from column name
iris.columns = iris.columns.str.replace(' ', '_')

iris.describe()

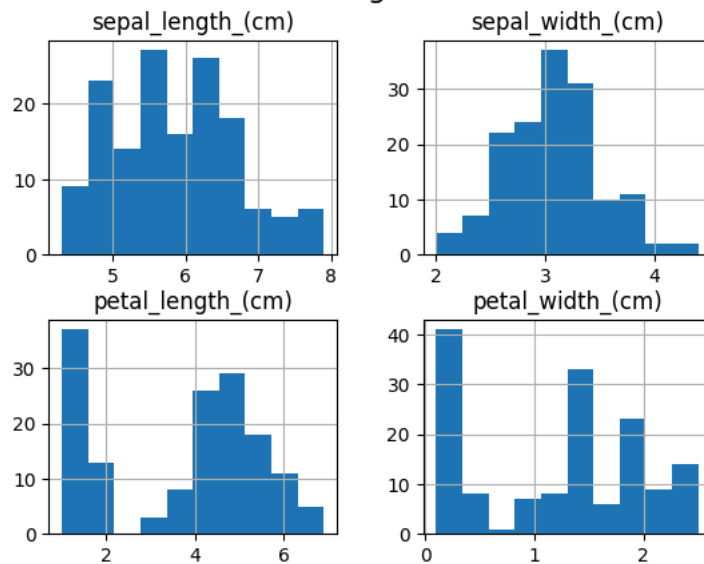
# The column 'species' is categorical, so let's check the frequency distribution
# for each category
print(iris['species'].value_counts())
```

```
species
setosa      50
versicolor  50
virginica   50
Name: count, dtype: int64
```

✓ Example 04: Pandas DataFrame **Visualization**

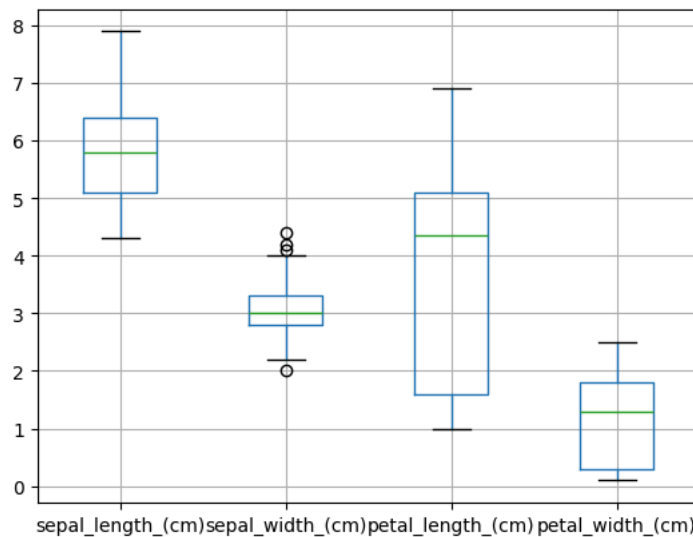
```
iris.hist() # plot histogram
plt.suptitle("Histogram", fontsize=16) # use suptitle to add title to all
plt.show()
```

Histogram



```
iris.boxplot() # plot boxplot
plt.title("Bar Plot", fontsize=16)
plt.show()
```

Bar Plot



Example 05: Multivariate Analysis

```
print(iris.groupby(by="species").mean())

# plot for mean of each feature for each label class
iris.groupby(by="species").mean().plot(kind="bar")

plt.title("Class vs Measurements")
plt.ylabel("mean measurement(cm)")
plt.xticks(rotation=0) # manage the ticks rotation
plt.grid(True)

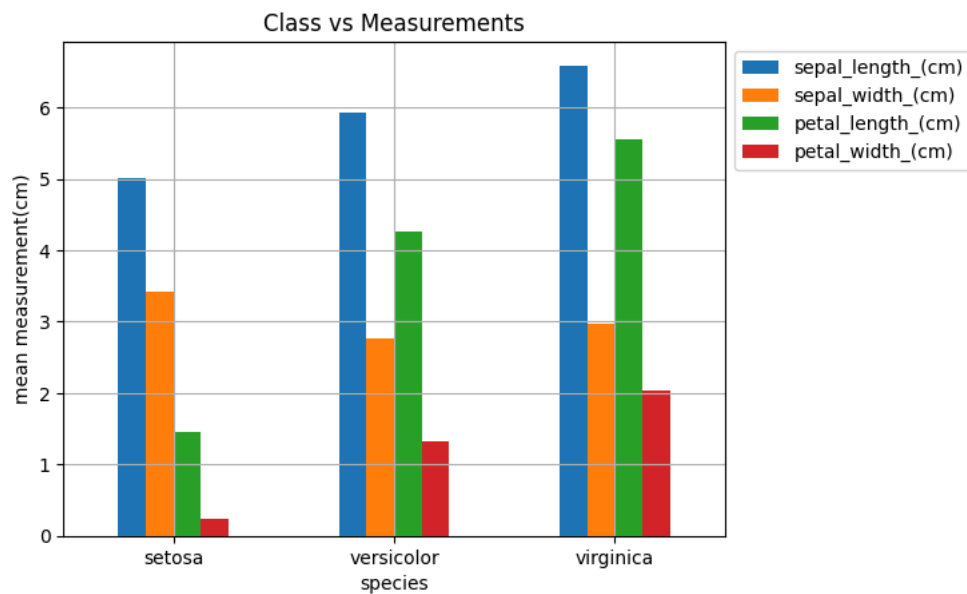
# Use bbox_to_anchor option to place the legend outside plot area
plt.legend(loc="upper left", bbox_to_anchor=(1,1))
```

```

species      sepal_length(cm)  sepal_width(cm)  petal_length(cm)  \
setosa        5.006            3.428            1.462
versicolor   5.936            2.770            4.260
virginica     6.588            2.974            5.552

species      petal_width(cm)
setosa        0.246
versicolor    1.326
virginica     2.026
<matplotlib.legend.Legend at 0x7d7c10692210>

```



Example 06: Correlation **Matrix**

```

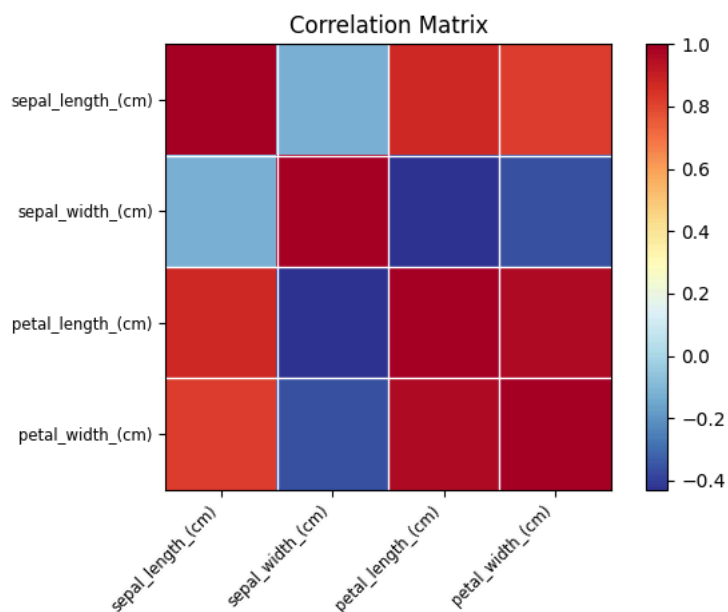
# create correlation matrix
corr = iris.corr(numeric_only=True)
print(corr)

import statsmodels.api as sm
sm.graphics.plot_corr(corr, xnames=list(corr.columns))
plt.show()

```

	sepal_length_(cm)	sepal_width_(cm)	petal_length_(cm)	\
sepal_length_(cm)	1.000000	-0.117570	0.871754	
sepal_width_(cm)	-0.117570	1.000000	-0.428440	
petal_length_(cm)	0.871754	-0.428440	1.000000	
petal_width_(cm)	0.817941	-0.366126	0.962865	

	petal_width_(cm)
sepal_length_(cm)	0.817941
sepal_width_(cm)	-0.366126
petal_length_(cm)	0.962865
petal_width_(cm)	1.000000

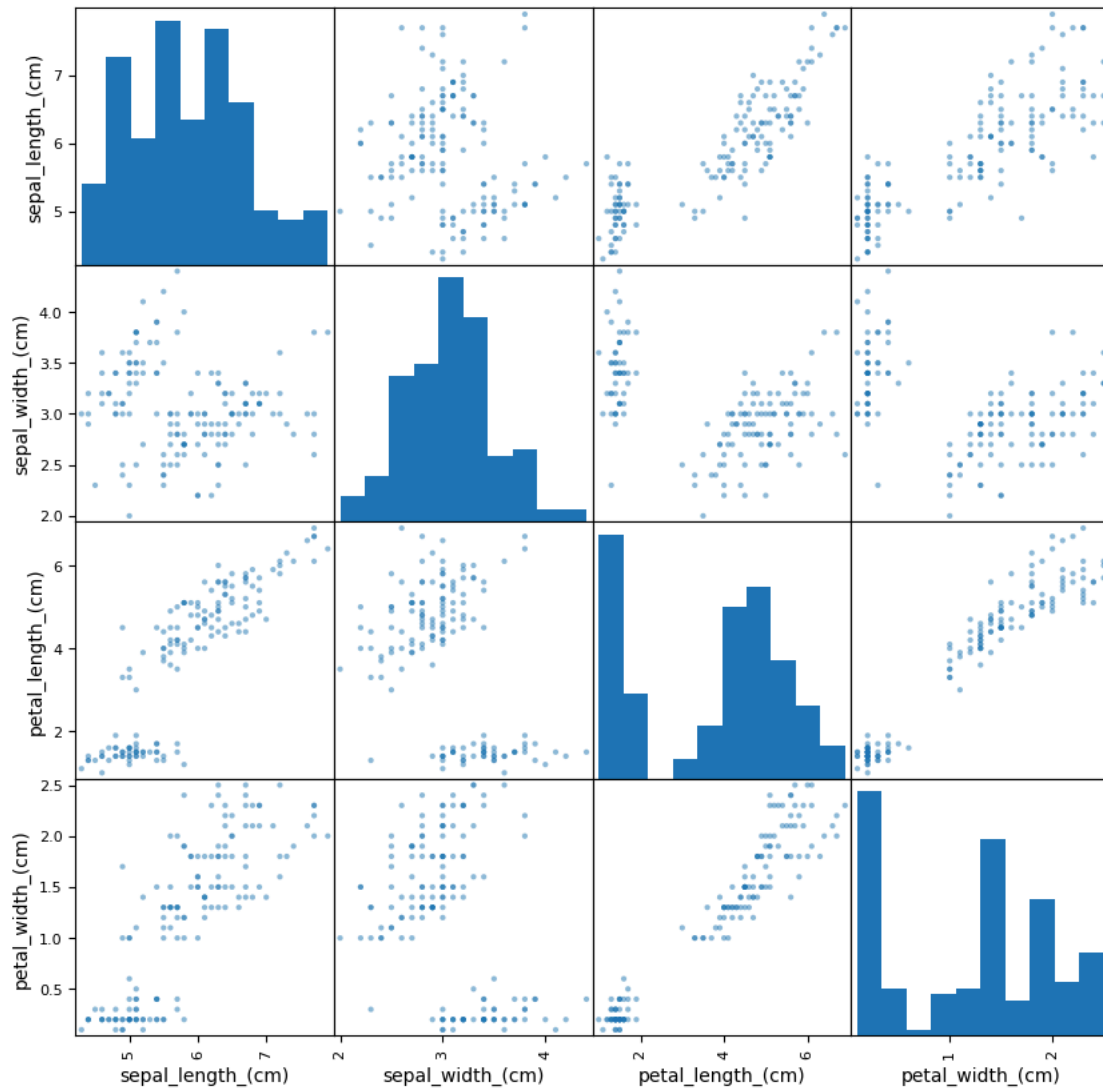


Example 07: Pair Plot

```
from pandas.plotting import scatter_matrix

scatter_matrix(iris, figsize=(10, 10))
plt.suptitle("Pair Plot", fontsize=20)
plt.show()
```

Pair Plot



Day 03: Matplotlib

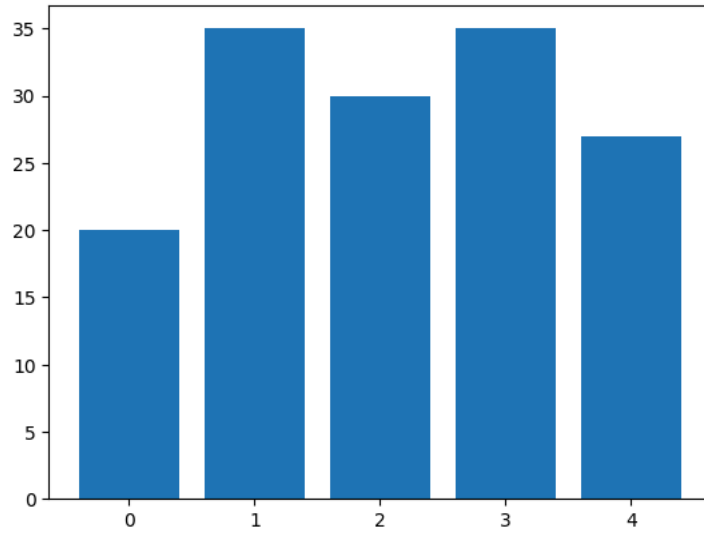
Example 01: Creating plot on **variables**

```
import numpy as np
from matplotlib import pyplot as plt

x = np.arange(5)
y = (20, 35, 30, 35, 27)

plt.bar(x, y)
plt.show()

plt.scatter(x, y)
plt.show()
```



Example 02: Creating plot on **dataframe**

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the Iris dataset
df = pd.read_csv('/content/iris.csv')

# Display the dataset
print(df.head())

# Histogram
df.hist()
plt.show()

# Line Graph
df.plot()
plt.show()

# Box Plot
df.boxplot()
plt.show()
```

Sepal.Length Sepal.Width Petal.Length Petal.Width Species